MILITARY SPECIFICATION

STEEL FORGINGS, ALLOY, STRUCTURAL, HIGH YIELD STRENGTH

(HY-130)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers HY-130 steel forgings intended primarily for use in combatant submarine hulls.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY
MIL-H-6875 - Heat Treatment of Steels (Aircraft Practice) Process for.

STANDARDS

MILITARY
MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.
MIL-STD-248 - Welding and Brazing Procedure and Performance Qulalification.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-1601 - Method for 5/8 Inch Dynamic Tear Testing of Metallic Materials.

PUBLICATIONS

MILITARY
NAVAL SEA SYSTEMS COMMAND
NAVSHIPS 0900-LP-005-5000 - Standard Procedures for Preproduction Testing Materials by the Explosion Bulge Test.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM A 370 - Mechanical Testing of Steel Products, Methods and Definitions for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.3).

3.2 Material.

3.2.1 All forgings shall be made from ingots or slabs which have been made by the same process used for production of the first article inspection forgings. The steel shall be fully killed and made to a fine grain practice.

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3.2.2 Billets and other material used in the production of forgings shall not be weld repaired.

3.2.3 Discard. Sufficient discard shall be taken from each ingot to ensure freedom from piping and undue segregation.

3.3 Forging process.

3.3.1 The original cross-sectional area of the ingot shall be at least three times the cross-sectional area of the main body of the forging. Palms, flanges, and other enlargements on forgings need not be reduced to the ratio of 3 to 1, but shall be reduced in a ratio of not less than 1.7 to 1. If bored ingots are used, the wall of the ingot shall be reduced to at least 1/2 of its original thickness. Where an upsetting operation is employed, or the forging expanded on a mandrel, the metal shall be worked to an extent not less than that indicated above, but in no fixed ratio between the cross-sectional area of the ingot and that of the forging.

3.3.2 Where the forgings are to be bored, the centerline of the ingot shall be either in a bored hole or in discarded metal.

3.4 Heat treatment. The forgings shall be quenched and tempered. The producer shall determine the detailed procedure for heat treating the forgings to meet the mechanical property requirements with the exception that the final austenitizing temperature shall be specified by the producer and shall not exceed 1650° fahrenheit (°F), and the tempering temperature shall be not less than 1000°F. Forgings shall be water quenched after tempering. The producer shall maintain a complete (austenitizing and tempering, temperatures, and times) record of the heat treatment, including reheat treatments given each forging and record the same on the certified report.

3.4.1 Heat treatment equipment and controls. Continuous or automatic heat treating equipment may be employed providing such equipment is capable of producing heat treated forgings to meet the requirements of this specification. Recording pyrometric equipment shall be employed to indicate the temperature of the forging during the heating and cooling cycles of heat treatment. Furnaces used in heat treatment of HY-130 forgings shall meet the quality assurance provisions of MIL-H-6875.

3.4.2 Reheat treatment. The manufacturer shall be permitted to reheat treat forgings which fail to meet tensile or impact requirements of this specification. All required tests originally performed on the failed forgings except chemical analysis and ultrasonic inspection (if previously performed), shall be repeated when the material is reinspected.

3.5 Chemical composition. The chemical analysis (heat and product) shall conform to the requirements of table I.

Table I - Chemical composition.

Element	Percent maximum unless a range is shown
Carbon	0.12
Manganese	0.60 - 0.90
Phosphorus	0.010
Sulfur	0.010
Silicon	0.20 - 0.35
Nickel	4.75 - 5.25
Chromium	0.40 - 0.70
Molybdenum	0.30 - 0.65
Vanadium	0.05 - 0.10
Residual elements	Maximum percent
Titanium	0.02
Copper	0.25

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3.5.1 Product analysis variations. On product analysis, the percentages of the elements shall be not over the upper limits or under the lower limits specified in table I by more than the amounts specified in table II. Where no product analysis variation is allowed, the limits for product analysis and heat analysis are the same.

Table II - Product analysis variations.

Element	Variations	
	Over the upper limit	Under the lower limit
Carbon	0.02	---
Manganese	0.05	0.05
Phosphorus	0.002	---
Sulfer	0.002	---
Silicon	0.02	0.02
Nickel	0.07	0.07
Chromium	0.04	0.04
Molybdenum	0.03	0.03
Vanadium	0.01	0.01

3.6 The material shall meet the requirements for tensile properties specified in table III.

Table III - Tensile properties (longitudinal and transverse).

Yield strength, 0.2 percent offset, lb/in^2	130,000 to 145,000
Ultimate tensile strength, lb/in^2	1/
Elongation in 2-inch (minimum percent) ASTM A370 standard .500 inch round specimen	15
Reduction of area, (minimum percent) ASTM A370 standard .500 inch round specimen	50

1/ Not required, to be recorded for information only.

3.7 Impact requirements. Unless otherwise specified (see 6.2), the forgings shall exhibit a minimum average Charpy V-notch impact energy absorption (longitudinal and transverse in plate type and cylinder forgings, tangential and longitudinal in mandrel expanded ring forgings) of 55 foot-pounds (ft-lbs), with no value below 40 ft-lbs. The average value for longitudinal and transverse $\left(\frac{L + T}{2}\right)$ are to be determined at 0°F ± 3°F and room temperature (70°F min). The average ft-lbs at room temperature shall not exceed by more than 15 ft-lbs the average at 0°F.

3.7.1 As an alternative to Charpy impact testing, transverse dynamic tear testing may be substituted at the option of the producer. At 0°F, 500 ft-lbs with 5/8-inch dynamic tear test specimen are required. For the 1-inch specimen, 4000 ft-lbs are required. Sampling and location of the dynamic tear test specimens shall be the same as required for Charpy impact tests except only two specimens shall be required for each lot.

3.8 Stress relief. Forgings shall not be stress relieved subsequent to final heat treatment.

3.9 Cleaning. Scale due to forging at elevated temperatures shall be removed to sound base metal so that visual, dimensional, and non-destructive evaluation can be satisfactorily accomplished.

3.10 Dimensions and tolerances. Forgings shall meet the dimensions and tolerances specified on the applicable drawings, (see 6.2). The responsibility for furnishing forgings that can be machined to the finished dimensions within the tolerances given and without further straightening shall rest with the contractor. Layout points, when required, shall be shown as such on the applicable drawings and shall be suitably incorporated in the forgings. Forgings outside of weight or dimensional tolerances shall be subject to rejection.

3.11 Soundness. Each forging shall be free of harmful defects as determined by visual examination, magnetic-particle testing, and ultrasonic inspection.

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3.11.1 Magnetic particle inspection. Unless otherwise specified (see 6.2), 100 percent of each forging's surface shall be magnetic particle inspected in accordance with 4.5.5. The forging shall be in the finished condition ready for shipment as specified. Forgings

3.11.2 Ultrasonic inspection. Each forging shall be ultrasonically tested for internal soundness. Any discontinuity whose reflection exceeds the calibration standard set forth in the table of MIL-STD-271 titled "Calibration hole size for longitudinal test" shall be cause for rejection of the forging.

3.12 Explosion bulge. Two crack starter and two bulge specimens are required for a complete series. The explosion bulge criteria of performance shall be in accordance with the following:

(a) Crack starter.
First shot
Crack starter bead must crack.
Percent reduction is thickness obtained for information only.
No piece shall be thrown out of material being tested.
No through thickness cracks shall be present.
No crack shall extend into hold down area.1/
Second shot
Percent reduction in thickness obtained for information only.
No piece shall be thrown out of material being tested.
Through thickness cracks are acceptable.
No cracks shall extend into hold down area.1/

(b) Bulge.
First shot
Percent reduction in thickness obtained for information only.
No piece shall be thrown out of material being tested.
No through thickness cracks shall be present.
No cracks shall extend into hold down area1/
Second shot
Percent reduction in thickness obtained for information, five percent reduction is expected.
No piece shall be thrown out of material being tested.
No through thickness cracks shall be present.
No crack shall extend into hold down area.1/
In the event that 7 percent reduction in plate thickness is obtained on both sides of the weld or 7-1/2 percent reduction in plate thickness is obtained on one side of the weld the performance is considered satisfactory and the third shot shall not be made provided the following conditions are met:
No piece shall be thrown out of material being tested.
Through thickness cracks are acceptable.
No cracks shall extend into the hold down area.1/
Third shot
Percent reduction in thickness obtained for information, seven percent is expected.
No piece shall be thrown out of material being tested.
Through thickness cracks are acceptable.
No cracks shall extend into hold down area.1/

1/ The bulge area is defined as that plate over the diehole (9-inch radius) plus the rounded outside corners (3-inch radius) plus 1/2 inch for a total circle diameter of 25 inches. The hold down area is defined as the area outside of this circle.

3.13 Defect repair. Defects may be removed by chipping, grinding, or other mechanical means, provided the involved area is well faired into the surrounding area and the design dimensions are not violated. Defects shall not be removed by arc cutting or oxygen cutting processes. Weld repair is prohibited unless specifically approved on a case basis by the command or agency concerned. The contour of the excavated area shall have a minimum radius of 3/8 inch. Repaired areas shall be magnetic particle inspected and meet the criteria of 3.11.1.

3.14 Marking. Each forging shall be permanently marked with the manufacturer's name or trademark, the designation HY-130, and a serial number which will positively identify the forging's part number, melt, and heat treated lot. Markings shall be placed in a location such that they are not machined off in finishing and in an area that is stressed least in service.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system requirements. The supplier shall provide and maintain an inspection system in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.1.2 Certification. Certification shall be provided for forgings as follows (see 6.2.2):

(a) Identification of the forging.
(b) Copies of certified test reports and inspection data setting forth results of examinations and tests. A copy of certification documents are to be sent to the consignee and the procuring activity, or as otherwise specified.
(c) Statement "Records are available covering heat number of the material used, processing of the forging, nondestructive testing, and heat treatment."

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall consist of the examination of 4.4.3 and tests specified in 4.5.

4.3.1 Sample for first article tests. The manufacturer shall provide enough two-inch thick plate material to provide the test data required by 4.3.2 and weldments as required in NAVSHIPS 0900-LP-005-5000.

4.3.2 Special instructions. Prior to beginning production, the manufacturer shall furnish the following information:

(a) Standard specification data (chemistry, physical properties, visual and ultrasonic inspection results from typical production parts).
(b) Charpy V-notch transition curves (longitudinal and transverse) with a minimum of five data points from minus 120°F to room temperature. A minimum of three specimens for each point is required and all individual values are to be reported. As an alternative, dynamic tear test transition curves (longitudinal and transverse) may be used. A minimum of five data points from minus 120°F to room temperature, and a minimum of two specimens for each point is required.
(c) Report of production line operations (such as melt practice, processing history, and heat treatment details).
(d) A copy of the weld procedure employed when explosion bulge plates are required.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests of 4.5.1, 4.5.2, 4.5.3, 4.5.4, and 4.5.6. The impact test shall be at only 0°F and 70°F for both the longitudinal and transverse specimens.

4.4.1 Lot definitions. Lot sizes are defined as follows:

(a) Chemical analysis - Each melt or heat of steel constitutes a lot.
(b) Mechanical tests - All forgings weighing less than 250 pounds produced from one heat or melt, heat-treated under similar conditions in the same heat-treating charge shall constitute a lot.
(c) Mechanical tests - Each forging weighing more than 250 pounds shall be considered a lot.
(d) Visual examination and inspection - Each forging shall be considered a lot.

4.4.2 Test samples.

4.4.2.1 Chemical or spectrographic analysis. Two samples shall be taken at random from each lot. Samples may be poured from the ladle in the form of a test block or drilled.

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from the forging proper, provided it is drilled in such a manner as not to impair the forging's usefulness. Where one forging represents the product of a heat, one chemical sample will suffice.

4.4.2.2 Mechanical properties. Sampling for mechanical properties shall be as follows:

(a) From each lot as defined by 4.4.1 (b) 25-percent of the forgings shall be tested for mechanical properties.
(b) From each lot as defined by 4.4.1 (c) each forging shall be tested for mechanical properties.

4.4.2.2.1 Location of mechanical test specimens. Unless otherwise specified (see 6.2), the location of test specimens shall be in accordance with figure 1, 2, or 3 as applicable.

4.4.2.3 Multiple forgings. When forgings are made and heat treated in multiple, such as two or more individual pieces are machined from a single heat-treated forging, specimens representing the composite forging shall be required. The composite forgings weight and size shall govern the lot definition and scheme of testing.

4.4.2.4 Marking of test specimens. The test specimens shall be marked to ensure positive identification of the lot being tested.

4.4.3 Visual and dimensional examination. Each forging shall be examined visually and dimensionally in accordance with 3.10 to assure conformance to drawings and freedom from defects harmful to the intended use.

4.5 Tests.

4.5.1 Chemical or spectrographic analysis. Samples selected in accordance with 4.4.2.1 shall be analyzed in accordance with a standard ASTM method or a method that will assure equally accurate results for conformance to tables I and II as applicable. The methods shall be correlated with National Bureau of Standards, standard reference materials when available to insure the validity of the method that is used as a control in chemical methods of analysis or for calibration in instrumental methods of analysis. The specification covering method of analysis used should be noted on certification documents.

4.5.2 Tensile tests. Tensile test specimens selected in accordance with 4.4.2.2 shall be tested in accordance with ASTM A370 to determine whether the material meets the requirements of table III.

4.5.3 Dynamic tear test. Dynamic tear test specimens selected in accordance with 4.4.2.2 shall be tested in accordance with MIL-STD-1601.

4.5.4 Charpy V-notch impact test. Charpy V-notch impact test specimens selected in accordance with 4.4.2.2 shall be tested for conformance with 3.7. A total of six specimens (three longitudinal and three transverse) from each lot shall be cooled by appropriate means to 0°F + 3°F in accordance with ASTM A370. Every precaution shall be taken to assure that the specimen has reached the temperature of the coolant, and the tongs used in handling the specimen shall be cooled with the specimen. Three specimens, from each lot, shall also be tested at room temperature (70°F minimum) after reaching a stabilized temperature.

4.5.5 Explosion bulge test. The explosion bulge tests shall be conducted and specimens fabricated in accordance with NAVSHIPS 0900-LP-005-5000. Specimens are to be tested with weld reinforcement in place.

4.5.6 Nondestructive evaluation (NDE). Magnetic particle and ultrasonic testing methods and procedures shall be in accordance with MIL-STD-271. NDE equipment, procedures, and personnel shall be qualified in accordance with MIL-STD-271.

4.6 Retests.

4.6.1 Tensile retest. If any forging fails to meet the tensile requirements, the forging may be reheat treated and retested in accordance with 4.5.2.

4.6.1.1 If the results on an original tensile specimen are within 1,000 pounds per square inch of the required yield strength, or within two percent of the required elongation or within two percent of the required reduction in area, a retest (specimen selected from an adjacent area from which the failed specimen was obtained) will be permitted.

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4.6.1.2 If the percentage of elongation or reduction of area of any tensile specimen is less than that specified in table III and any part of the fracture is outside the gage length, or within the gage length and less than 25 percent of the gage length from either datum point, another specimen from and adjacent area from which the failed specimen was obtained may be selected for a retest.

4.6.2 Impact retest. In the event that only one of the impact test specimens (Charpy or dynamic tear) representing a forging is below the specified minimum for a single impact test value, two additional specimens may be tested with both specimens meeting specification requirements (40 ft-lbs for Charpy V-notch or 500 ft-lbs for dynamic tear). The forging may be reheat treated and retested in accordance with 4.5.3 or 4.5.4.

4.6.3 Defective specimen. If any test specimen shows defective machining or obvious lack of continuity of metal, it may be discarded and another selected from an adjacent area from which the failed specimen was obtained may be selected for a retest.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing, and marking. Forgings, cleaned and preserved as specified (see 3.9), shall be packed level A or C, as specified (see 6.2), in wood boxes or on skid systems and marked in accordance with MIL-STD-163.

6. NOTES

6.1 Intended use. The HY-130 steel forgings covered by this specification are intended for combatant submarine hull use.

6.1.1 This steel may be used in surface ship construction or other critical structural applications where a weldable, notch tough, high strength material is required.

6.1.2 This steel can also be used to fabricate welded pressure vessels and other machinery items of critical use where an as-welded, notch-tough, high yield strength steel is required.

6.1.3 The use of HY-130 steel in fabricated structure or equipment entails much more than a material specification and caution is advised in the areas of welding, fabrication, and nondestructive evaluation.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) When Charpy V-notch impact absorption is other than specified (see 3.7).
(c) Dimensions and tolerances required (see 3.10).
(d) When magnetic particle inspection is other than 100 percent (see 3.11.1).
(e) When location of test specimens is other than specified (see 4.4.2.2.1).
(f) Level of packing required (see 5.1).

6.2.2 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

	Specification paragraph	Data requirements	Service	Applicable DID	Options
(a)	4.1.1	Inspection System Program Plan	SH	DI-R-4803	---
(b)	4.1.2	Certification Data/Report	SH	UDI-A-23264	---

(Copies is DID'S required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

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6.3 First article inspection.

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
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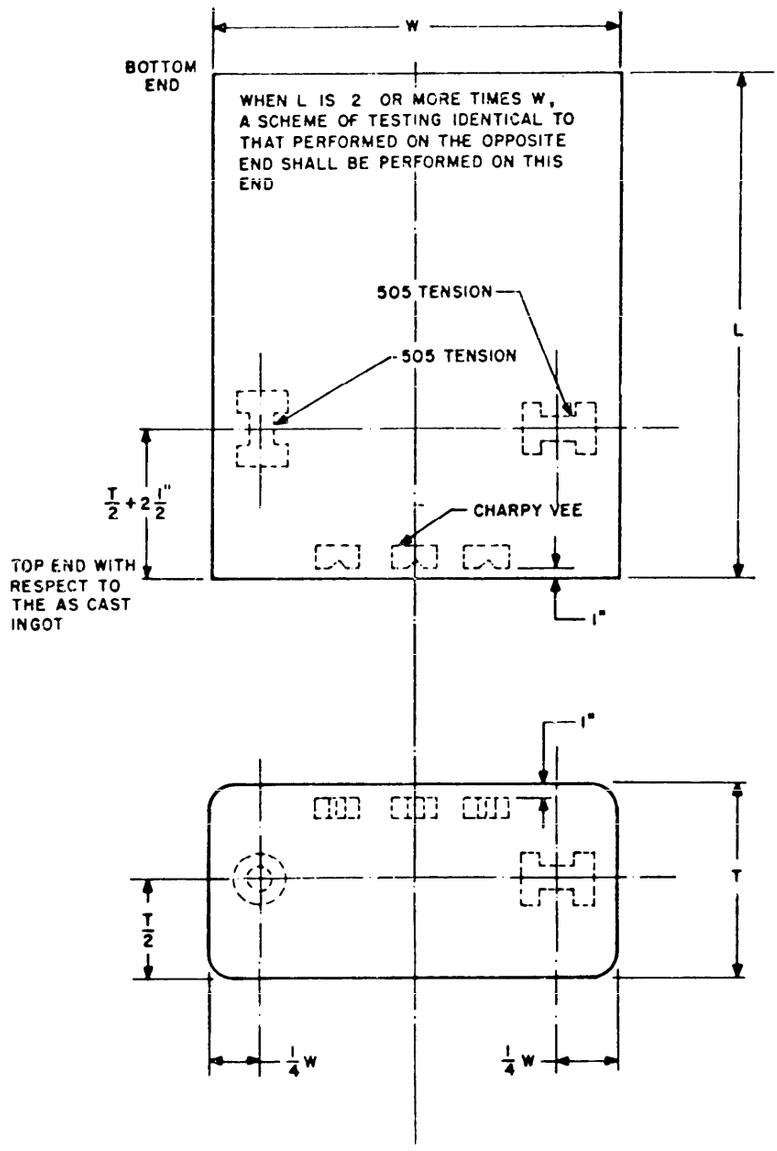


Figure 1 - Schematic diagram of test specimen location for forgings "rectangular-like" in cross section

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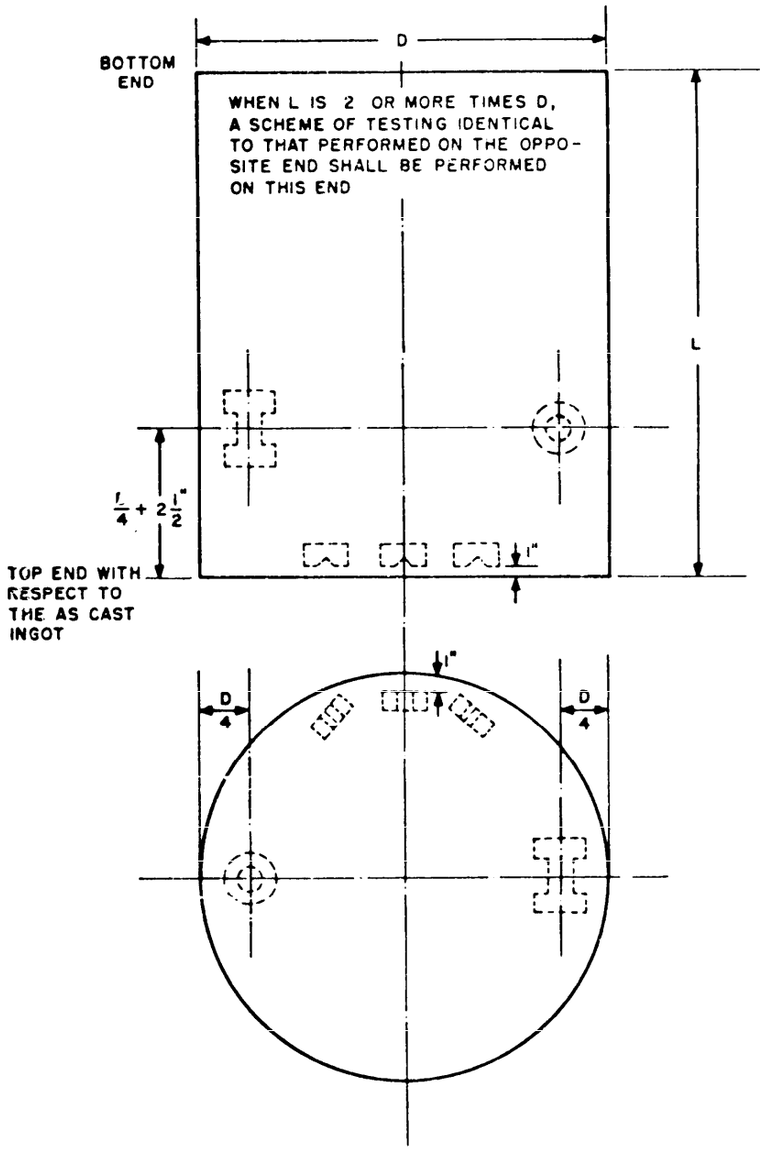


Figure 2 - Schematic diagram of test specimen location for forgings of solid circular cross section

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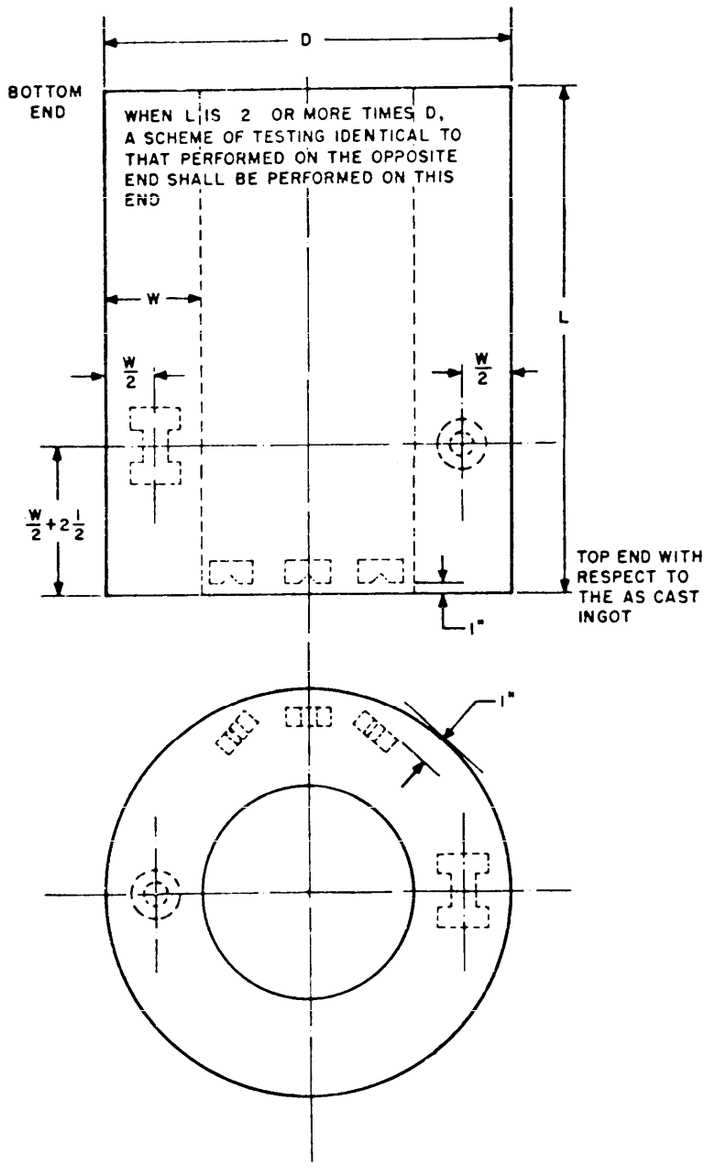


Figure 3 - Schematic diagram of test specimen location for forgings of circular cross section.

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DOCUMENT IDENTIFIER AND TITLE

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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